AI Meeting Assistant

Mini Assessment Work

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Topic	AI Meeting Assistant – Automated Meeting Summarization and Action-Item Extraction Using Generative AI

1. Introduction

The **AI Meeting Assistant** is a Generative AI-based application that converts meeting conversations or transcripts into a short, organized and easy-to-understand summary. It also identifies important action items, responsible persons and deadlines. The main goal is to reduce the time required to review long meeting discussions and make follow-up tasks clear.

2. Problem Statement

After a meeting, participants may forget important decisions, assigned tasks or deadlines. Reading the complete transcript again can be time-consuming. A simple AI system is therefore needed to automatically extract the most useful information from meeting conversations.

3. Objectives

- Summarize a meeting in a concise format.
- Identify important tasks and action items.
- Extract the person responsible for each task.
- Identify deadlines when they are mentioned.
- Present the output in a clear and structured format.

4. Proposed Solution

The user provides meeting text or a transcript to the application. The Generative AI model understands the conversation and produces a structured result containing a meeting summary, decisions and action items. The output can be displayed through a simple web interface.

5. Working of the System

Step 1 – Input: The user enters or uploads a meeting transcript.

Step 2 – Pre-processing: The application prepares the text for the AI model.

Step 3 – AI Analysis: A Generative AI model understands the conversation and identifies key information.

Step 4 – Information Extraction: The model generates summary, tasks, people and deadlines.

Step 5 – Output: The results are shown in a simple and readable interface.

6. Simple System Flow

Meeting Transcript
↓
Text Processing
↓
Generative AI Model
↓
Summary + Action Items + Person + Deadline
↓
Simple Web Interface

7. Technologies Used

Technology	Purpose
Python	Main programming language
Streamlit	Simple web interface
Gemini / OpenAI API	Generative AI processing
Prompt Engineering	Guide the AI to produce structured results
Text / PDF processing	Read meeting transcript files

8. Example

Input: “Priya will complete the frontend by Wednesday. Arun will connect the database by Thursday. The team will test the complete application on Friday.”

Person	Action Item	Deadline
Priya	Complete frontend	Wednesday
Arun	Connect database	Thursday
Team	Test the application	Friday

Generated Summary: The team discussed frontend completion, database integration and final testing, with tasks assigned to Priya, Arun and the team.

9. Key Generative AI Concept

Generative AI and Prompt Engineering: The project uses a Large Language Model (LLM) to understand natural-language meeting conversations and generate new, structured text. A carefully designed prompt instructs the model to return the information in a consistent format, such as summary, decisions and action items.

10. Expected Output

Section	Information Generated
Meeting Summary	Short overview of the discussion
Key Decisions	Important decisions made during the meeting
Action Items	Tasks that need to be completed
Responsible Person	Person assigned to each task
Deadline	Due date or time when mentioned

11. Advantages

- Saves time by avoiding manual review of long transcripts.
- Makes responsibilities and deadlines easy to identify.
- Reduces the chance of missing important follow-up tasks.
- Provides a simple and user-friendly way to organize meeting information.

12. Future Enhancements

- Add voice-to-text for direct audio meeting input.
- Send reminders for pending action items.
- Integrate with calendars and task-management applications.
- Support multiple languages.

13. Conclusion

The AI Meeting Assistant is a simple and practical Generative AI project that converts unstructured meeting conversations into useful, organized information. By combining Python, a simple interface and an LLM, the system can help users quickly understand what was discussed and what needs to be done next. The project demonstrates a real-world application of Generative AI without requiring complex model training.

Mini Assessment Summary: Input meeting transcript → Generative AI analysis → Summary + decisions + action items → Clear structured output.